

Intermediate-phase engineering via dimethylammonium cation additive for stable perovskite solar cells

Received: 16 October 2018

Accepted: 7 October 2022

Published online: 1 December 2022

 Check for updates

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Achieving the long-term stability of perovskite solar cells is arguably the most important challenge required to enable widespread commercialization. Understanding the perovskite crystallization process and its direct impact on device stability is critical to achieving this goal. The commonly employed dimethyl-formamide/dimethyl-sulfoxide solvent preparation method results in a poor crystal quality and microstructure of the polycrystalline perovskite films. In this work, we introduce a high-temperature dimethyl-sulfoxide-free processing method that utilizes dimethylammonium chloride as an additive to control the perovskite intermediate precursor phases. By controlling the crystallization sequence, we tune the grain size, texturing, orientation (corner-up versus face-up) and crystallinity of the formamidinium (FA)/caesium (FA)_yCs_{1-y}Pb(I_xBr_{1-x})₃ perovskite system. A population of encapsulated devices showed improved operational stability, with a median T80 lifetime (the time over which the device power conversion efficiency decreases to 80% of its initial value) for the steady-state power conversion efficiency of 1,190 hours, and a champion device showed a T80 of 1,410 hours, under simulated sunlight at 65 °C in air, under open-circuit conditions. This work highlights the importance of material quality in achieving the long-term operational stability of perovskite optoelectronic devices.

With over 25% power conversion efficiencies (PCEs)¹, organic–inorganic metal halide perovskite solar cells have now reached a laboratory performance suitable for commercial applications. Although important performance breakthroughs have been achieved since

their humble beginnings^{2–6}, their stability under operating conditions has yet to match that of state-of-the-art silicon-based solar cell technology. Recently, a number of research groups have demonstrated the ability of perovskite solar cells to pass a selection of tests from

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existing International Electrotechnical Commission (IEC) qualification standards for terrestrial photovoltaic modules such as IEC 61215. However, this design qualification and test approval does not guarantee 25 years of operational stability. In particular, it is important to understand all types of degradations that may occur with this new solar absorber material during real-world use, which may not be tested for in IEC 61215 (refs. 7–10). Considerable advances have been made by mitigating thermal, structural, moisture-related and light-related instabilities found in early perovskite compositions. Moreover, these stability improvements have been enabled by enhanced encapsulation methods, ionic-liquid additives and Lewis acid–base passivation and by employing inherently more stable hole- and electron-accepting layers^{11–13}. Furthermore, a dramatic increase in thermal stability was achieved by replacing the volatile A-site cation, methylammonium (CH_3NH_3^+ ; MA), with a more thermally stable cation, FA ($\text{CH}(\text{NH}_2)_2^+$). Although neat FAPbI_3 forms in the desired cubic crystal structure at elevated temperatures ($>170^\circ\text{C}$), once cooled to room temperature, it suffers from structural instability, where the black 1.47 eV photoactive three-dimensional (3D) corner-sharing ABX_3 phase reverts to a yellow wide-bandgap face-sharing non-perovskite-phase crystal structure. Hence, a second cation, notably caesium (Cs^+), is often used to facilitate structural stabilization of the black α -phase of FA-based perovskites without the use of the unstable MA cation^{14–19}.

In this work, we focus on an often-overlooked aspect of perovskite stability: crystalline quality rather than composition. In recent years, the metal halide perovskite field has converged towards a simple and easy perovskite fabrication route²⁰ consisting of a dimethyl-formamide/dimethyl-sulfoxide (DMF/DMSO) solvent system along with an ‘antisolvent’ step, such as with chlorobenzene, toluene, diethyl ether or anisole ($\text{C}_6\text{H}_5\text{Cl}$, C_7H_8 , $(\text{C}_2\text{H}_5)_2\text{O}$ or $\text{CH}_3\text{OC}_6\text{H}_5$), to quench the perovskite growth. Although this method is unlikely to be industry compatible, it allows for facile fabrication of laboratory-scale prototypes, while providing an acceptable level of film reproducibility and uniformity for various perovskite compositions. During the spin-coating process, the antisolvent step triggers rapid nucleation of the perovskite/DMSO complexes, forming a series of intermediate phases, and extracts a large fraction of the original high-boiling-point solvents²¹. A subsequent annealing step provides the thermodynamic conditions allowing a 3D corner-sharing perovskite to fully form. However, these DMSO/lead-halide and DMSO/perovskite complexes, such as DMSO-PbX_2 and $(\text{DMSO})_2\text{-PbX}_2$, may lead to residual DMSO solvent being trapped within the film due to its strong molecular interaction with perovskite or PbX_2 and the high boiling point, and are just one of a whole range of possible precursor phase routes that could be explored and that could lead to a higher-quality perovskite^{22,23}. The antisolvent quenching approach promotes rapid heterogeneous nucleation with a high density of nucleation centres, leading to a short crystal growth period. This technique results in materials with relatively small grain sizes, polycrystalline films and a relatively high density of structural and energetic disorder. These defects tend to result in a higher trap-state density and shorter carrier diffusion lengths compared to lengthier homogeneous nucleation and growth methods, such as single-crystal growth techniques^{24,25}. Lastly, smaller grain sizes result in an increase in grain boundaries, leading to a higher surface-defect density. Structural defects located at the grain boundary have been linked to the promotion of moisture-induced degradation pathways²⁶. Even though these films operate very well initially in photovoltaic devices, improving the crystal quality towards a ‘single-crystal-like’ perovskite film should represent a step towards enhancing the long-term stability of perovskite films and devices.

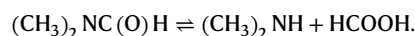
We have studied the precursor solution chemistry^{27,28} and observed the formation of dimethylamine ($(\text{CH}_3)_2\text{NH}$) and formic acid (HCOOH) in a solution via the hydrolysis of DMF. For the inorganic perovskite CsPbI_3 , we and others have shown that the presence of dimethylammonium iodide (DMAI), following hydrolysis of DMF via the addition of HI, results in DMA^+ being incorporated into the ‘now

hybrid’ perovskite $\text{DMA}_x\text{Cs}_{1-x}\text{PbI}_3$ (refs. 29,30). We have also observed that the colloidal chemistry of the $\text{FA}_y\text{Cs}_{1-y}\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ precursor solution changes over time when hydrohalic acids are added to accelerate this decomposition reaction^{28,31}. These hydrohalic acid additives vary the lead polyhalide colloid density in the precursor solution. We interpreted these results to indicate that these colloids, composed of the lead coordination compounds $[\text{PbX}_3]^-$, $[\text{PbX}_4]^{2-}$, $[\text{PbX}_5]^{3-}$ and $[\text{PbX}_6]^{4-}$, serve as nucleation centres that impact the crystallization kinetics of the perovskite formation, resulting in a change in texture, grain size and defect density^{31,32}. We showed that a precursor solution aged with hydrohalic acid additives resulted in highly crystalline and oriented perovskite grains. However, this method had some downfalls, where an ‘under-aged’ solution led to lower crystallinity, smaller grain sizes and lower charge-carrier mobilities. By contrast, an over-aged solution suffered from poor perovskite coverage across the substrate.

Here, we show that a key component of the acid-aged perovskite solutions is the dimethylamine generated from the decomposition of DMF. With this insight, we introduce a new controllable method of precursor phase engineering with the addition of dimethylammonium halide (DMAX), and specifically dimethylammonium chloride ($(\text{CH}_3)_2\text{NH}_2\text{Cl}$; DMAI). By utilizing DMAI as a crystallization agent, we can gradually transition via the hexagonal face-sharing perovskite polytypes (2H, 4H, 6H) towards a 3C corner-sharing perovskite, resulting in highly crystalline and textured thin films with a face-up perovskite unit cell orientation. Finally, we study the impact of improved material quality on the long-term stability of the FA-caesium-based perovskite using three distinct fabrication methods: the state-of-the-art DMF/DMSO processing route, the DMF/acid processing route and our new DMF/DMAI processing route. Our findings establish links between the details of fabrication methods, namely the ink formulation in concert with the detail of the processing route chosen to manipulate material quality and, ultimately, device stability.

Degradation of DMF

It has been shown that prolonged exposure to air and/or water degrades DMF to the secondary amine (dimethylamine) and formic acid^{28,33}:



Furthermore, in the presence of strong acids, such as HI or HBr, this process is accelerated, where hydrohalic acids act as catalysts for this reaction. We performed a time-of-flight secondary ion mass spectrometry (TOF-SIMS) measurement and detected residual amounts of DMA^+ in the perovskite film after a 185°C , 90 min annealing step in an oven for a film prepared from a DMF precursor solution aged for 48 h with hydrohalic acid additives (Supplementary Fig. 1). Hence, we confirm that these hydrohalic acids in water (57 wt% HI in H_2O ; and/or 48 wt% HBr in H_2O) cause fast hydrolysis of DMF while simultaneously providing a source of halide, causing an unintended and uncontrolled addition of DMAX salt, $(\text{CH}_3)_2\text{NH}_2^+\text{X}^-$, into the perovskite solution.

In Fig. 1, we show the impact of the hydrolysis reaction of DMF on the morphology, crystallographic properties and stability of halide compositional segregation under illumination of the $\text{FA}_y\text{Cs}_{1-y}\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ perovskite system. With the addition of hydrohalic acids, we see a change in the morphology of the films with the domains and apparent grains observed with scanning electron microscopy (SEM) changing in size from a few hundred nanometres to several micrometres (Fig. 1b,c), commensurate with the scattering measurements³¹. Furthermore, we study perovskite films based on a wide-bandgap (1.7–1.8 eV) $\text{FA}_y\text{Cs}_{1-y}\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ perovskite precursor, suitable for perovskite-on-silicon tandem solar cells or as the front cell in all-perovskite tandem cells^{14,34,35}. In Fig. 1d,e, we display wide-angle X-ray scattering measurements, where we observe that the control DMF/DMSO route results in concentric Debye–Scherrer rings indicative of polycrystalline thin films with a low degree of texture. On the other hand, we observed a large

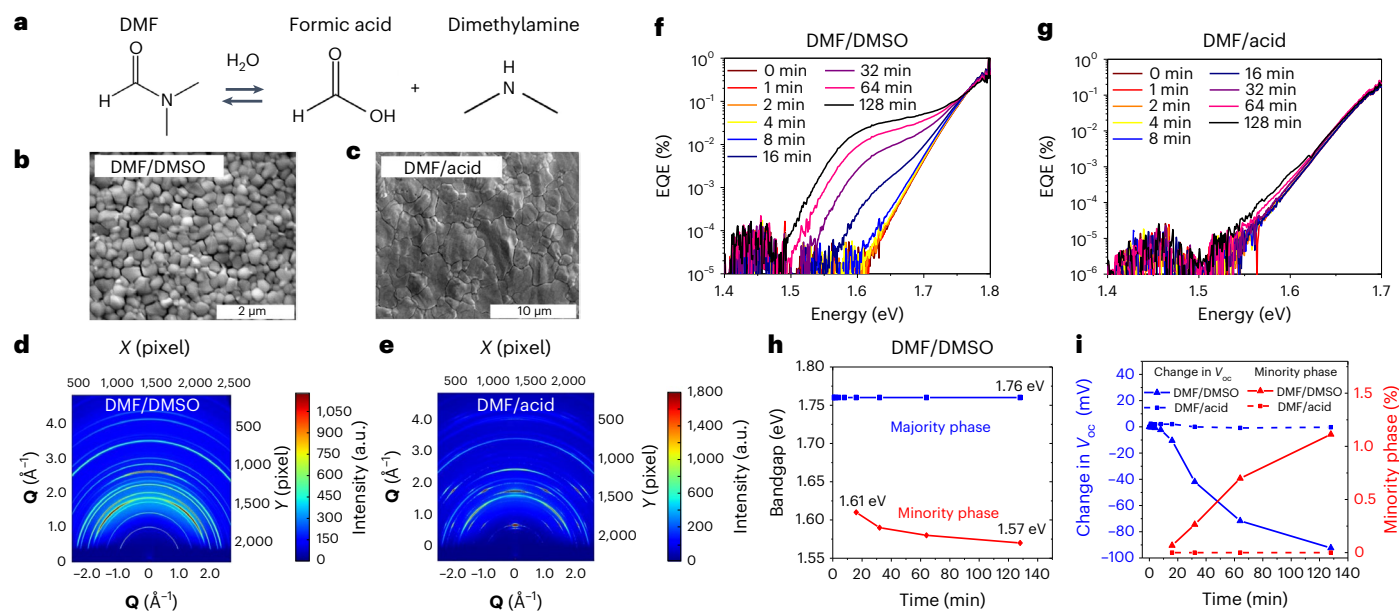


Fig. 1 | Impact of hydrohalic acids on the morphology, crystal quality and electronic disorder of the $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{I}_{0.6}\text{Br}_{0.4})_3$ perovskite film. **a**, Schematic showing the degradation of DMF into dimethylamine and formic acid in the presence of water. The chemical equation of the DMF hydrolysis reaction, where DMF is the reactant and dimethylamine and formic acid are the products, is $(\text{CH}_3)_2\text{NC(O)H} \rightleftharpoons \text{HCOOH} + (\text{CH}_3)_2\text{NH}$. **b,c**, SEM images of a $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{I}_{0.6}\text{Br}_{0.4})_3$ perovskite film prepared with a DMF/DMSO solvent mixture (**b**) and with hydrohalic acid (HI/HBr) additives (**c**). **d,e**, Wide-angle X-ray scattering images of $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{I}_{0.6}\text{Br}_{0.4})_3$ perovskite films deposited on a FTO substrate, where **Q** is defined as the scattering vector. **f,g**, EQE, measured via Fourier transform

photocurrent spectroscopy for a solar cell with a DMF/DMSO perovskite film (**f**) and a DMF/acid perovskite film (**g**). For the EQE measurements, the devices were non-encapsulated and illuminated by a xenon lamp simulated full-spectrum AM 1.5G, 100 mW cm^{-2} irradiance at open circuit in air (relative humidity, ~45%) at room temperature. **h**, Quantification of the bandgap of the mixed-halide majority phase and the iodine-rich minority phase as a function of irradiation time, estimated from the EQE spectra. **i**, Quantification of the V_{oc} loss due to the minority phase segregation, and the percentage of the minority phase as a function of irradiation time. a.u., arbitrary units.

variation in intensity across the Debye–Scherrer rings for the DMF/acid route, indicative of a highly textured film. In Supplementary Figs. 2–4, we present the in situ crystallization wide-angle X-ray scattering measurements for both fabrication methods, plotting the scattering intensity versus time, revealing that the DMF/acid films crystallize at a substantially slower rate compared to the DMF/DMSO films, even though they crystallize at a higher temperature (185°C versus 100°C).

It is often observed that mixed-halide perovskite materials with high bromide content suffer from halide segregation³⁶, in which localized iodide- and bromide-rich perovskite domains form, when the films are exposed to light. Here, we performed external quantum efficiency (EQE) measurements, over a prolonged period of light exposure, of solar cells fabricated with perovskite films processed from these two different routes to assess the relative rate of halide segregation (Fig. 1f,g)³⁷. The steepness of the sub-bandgap absorption onset gives information about the electronic disorder in the absorber layer. This is often quantified as the exponential gradient, termed the Urbach energy. The pristine DMF/DMSO devices (at 0 min) have an Urbach energy of 16.0 meV, while the device with the DMF/acid perovskite showed a lower Urbach energy of 13.8 meV (Supplementary Figs. 5 and 6). Surprisingly, although the initial precursor salt composition was identical for the DMF/DMSO and DMF/acid perovskite films, we found a strong time-dependent variation in electronic disorder between these two films. We follow our previously published approach in modelling the perovskite as being made of two components: a majority phase and a phase-segregated minority phase, which we describe in the Supplementary Information³⁷.

We observed a sharp differential between the bandgap of the majority phase ($E_{\text{g,majority}}$) of 1.76 eV and the bandgap of the minority phase ($E_{\text{g,minority}}$) of 1.57 eV after 128 minutes of 1 sun illumination for the DMF/DMSO film. However, the DMF/acid films showed little to no variation in bandgap over the same period (Fig. 1f). Lastly, Supplementary

Figs. 7–10 show the TOF-SIMS tomography of pristine and aged perovskite devices with the DMF/DMSO and DMF/acid perovskite films. The devices were aged under 1 sun illumination at the maximum power point for 24 hours. This measurement allowed us to assess the inhomogeneities of the FA/Cs A-site cations and the I/Br halides. After ageing, we observed a stronger compositional variation for the DMF/DMSO perovskite films than the DMF/acid films, consistent with the improved compositional stability of the DMF/acid-processed films.

From the measurements of the light-soaked devices, we estimated that the minority phase bandgap represents ~1.1% of the material after 120 minutes of illumination for the DMF/DMSO films, resulting in over 90 mV estimated loss in the open-circuit voltage, V_{oc} . On the other hand, we observe only a slight increase in Urbach energy under illumination for the DMF/acid devices, but this increase remains well below the thermal energy, $k_{\text{B}}T$, where k_{B} is Boltzmann's constant and T is temperature. The absorption edge does not exhibit any discernible characteristic shoulder, suggesting that compositional disorder increased only slightly; however, no distinct minority phase was formed. The corresponding open-circuit voltage loss due to this degree of segregation can be calculated to be less than 1 mV. As previously reported, we attribute this enhanced segregation stability to a higher level of ordering within the crystal lattice and/or larger grain sizes³⁸. One possible hypothesis is that the improvement in crystal quality lowers the formation of native point defects such as vacancies, such as $\text{V}_{\text{FA,Cs}}$, V_{Pb} and $\text{V}_{\text{I,Br}}$; interstitials, such as FA_{i} , Cs_{i} , Pb_{i} and I_{i} , Br_{i} ; and anti-site occupations, such as FA_{Pb} , Cs_{Pb} , $\text{FA}_{\text{I,Br}}$, $\text{Pb}_{\text{FA,Cs}}$, $\text{Pb}_{\text{I,Br}}$, $\text{I}_{\text{FA,Cs}}$, $\text{Br}_{\text{FA,Cs}}$ and I_{Pb} , Br_{Pb} that can promote ionic motion within the perovskite crystal³⁹. It has also been shown that the rate of halide segregation increases with increased non-radiative or trap-assisted recombination^{40,41}. This may be due to the hole-trapping process being responsible for liberating the mobile iodine diffusion, which is a key part of halide segregation.

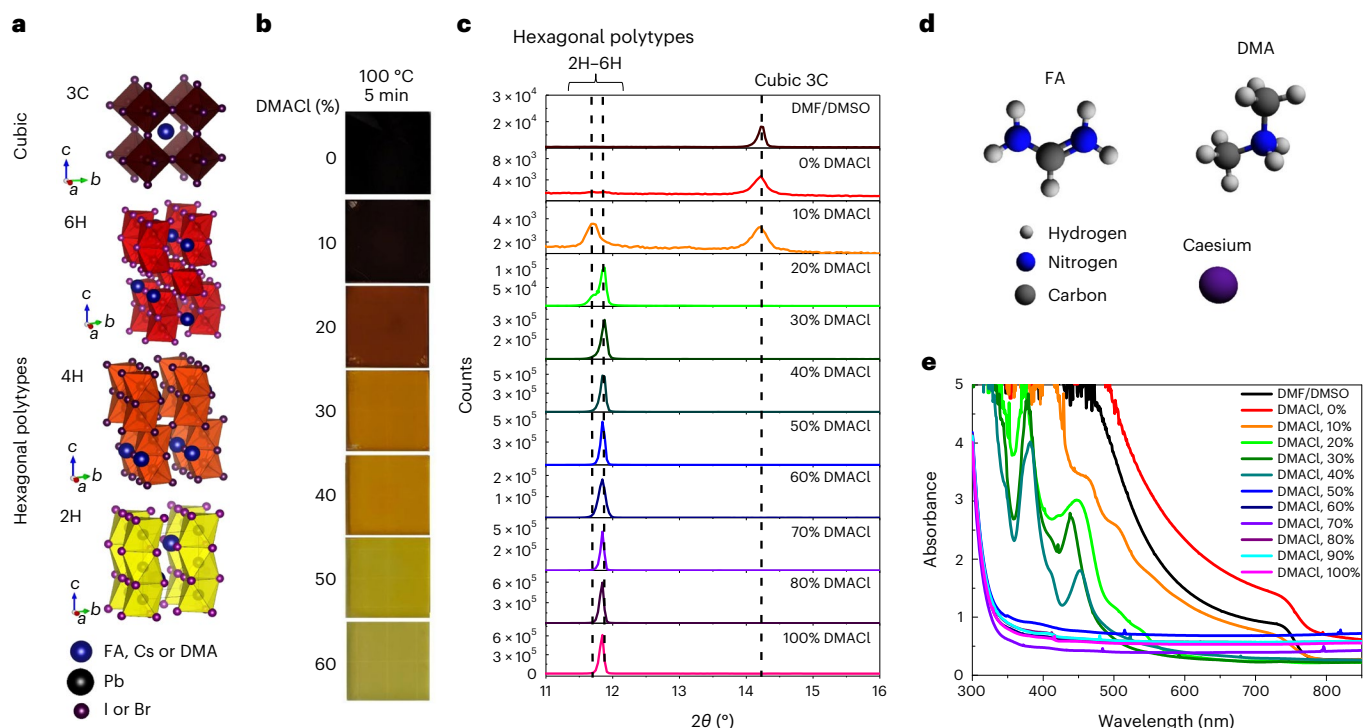


Fig. 2 | Impact of DMAcI on the intermediate phases of $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$. **a**, Crystal structure of a unit cell of the hexagonal lead halide perovskite polytypes—2H, 4H and 6H—showing PbI_6 octahedra connectivity, along with a 3C cubic perovskite structure. **b**, A series of photographs of $\text{DMAcI}(\text{FA}_{0.83}\text{Cs}_{0.17})_{1-x}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3\text{Cl}_x$ perovskite films spin-coated with various amounts of DMAcI

additive, where percentages are expressed in excess amounts with respect to lead. **c**, A series of X-ray diffraction patterns of the corresponding thin films showing the intermediate precursor phases in the form of hexagonal polytypes. **d**, A schematic of the A-site cations used in the perovskite crystallization process. **e**, UV-visible absorbance spectra of the corresponding thin films.

Formation of intermediate phases via DMAcI

Intermediate, or precursor, phases commonly occur during the formation of polycrystalline metal halide perovskite thin films⁴². Although clearly influential, they are often an overlooked aspect of the complex crystallization process driven by thermodynamics. We observed that films prepared with a precursor solution aged with hydrohalic acids crystallized via distinct intermediate phases. We postulate that the source of these specific intermediate phases, and thus control of the formation process, is the unintentional presence of DMA^+ . To test this theory, we prepared perovskite thin films from precursor solutions that contained increasing concentrations of DMAcI salts. Consistent with our hypothesis, we observed that this addition of DMAcI induces the formation of perovskite precursor phases, which allows for crystallization to proceed via the formation of hexagonal perovskite polytype (2H, 4H and 6H) intermediate phases, which we illustrate in Fig. 2. This class of perovskite polytypes is composed of a mixture of corner-sharing (cubic) and face-sharing (hexagonal) octahedral connections assembled in various ratios. Members of this family of perovskites are commonly found in perovskite oxides, but have also been reported to occur in metal halide perovskites^{43,44}. Here, we identify the $\text{DMAcI}(\text{FA}_{0.83}\text{Cs}_{0.17})_{1-x}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3\text{Cl}_x$ perovskite system as part of this polymorph class of material, where the proportion of cubic to hexagonal octahedral connections increases when exposed to high temperatures (>140 °C; Supplementary Fig. 11), resulting in a phase transition from 2H to 4H to 6H face-sharing hexagonal frameworks, to eventually converge towards a fully 3C corner-sharing cubic perovskite.

It has been reported that $\text{DMAcI}(\text{FA}_{0.83}\text{Cs}_{0.17})_{1-x}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3\text{Cl}_x$ forms in various hexagonal face-sharing crystal arrangements^{45,46}. For example, $\text{DMAcI}(\text{FA}_{0.83}\text{Cs}_{0.17})_{1-x}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3\text{Cl}_x$ crystallizes in a 2H-hexagonal polytype with space group $P6_3/mmc$, comprising chains of face-sharing $[\text{PbI}_6]^{4-}$ octahedra separated by chains of DMA^+ cations⁴⁶; whereas $\text{DMAcI}(\text{FA}_{0.83}\text{Cs}_{0.17})_{1-x}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3\text{Cl}_x$

and $\text{DMAcI}(\text{FA}_{0.83}\text{Cs}_{0.17})_{1-x}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3\text{Cl}_x$ both crystallize in a 4H-hexagonal perovskite polytype. These 4H perovskite polytypes are formed with Pb_2X_9 dimers of face-sharing octahedra, which are then connected to one another via corner-sharing octahedra⁴⁷. Figure 2 shows the intermediate phases occurring in the solvent-dried films, prior to the conversion to the fully crystalline 3C perovskite form. After a 5 minute, 100 °C anneal, we observe a colour change, along with the appearance of the 2H to 6H hexagonal polytype phase, identified by X-ray reflections between $2\theta \approx 11.6^\circ$ and 11.8° , where 2θ is the angle between transmitted beam and reflected beam. Moreover, we observe that with $\geq 20\%$ excess DMAcI (with respect to the Pb or combined FA + Cs content), we fully suppress the formation of the 3C cubic $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$ perovskite when heated at or below 100 °C. We assume that most of the DMA solvent has been removed from the film during this 100 °C annealing process, but most of the DMAcI remains incorporated within the crystalline polymorphs. With a subsequent anneal at higher temperatures, we assume that most of the DMAcI (the most volatile remaining component) is driven out of the film, until only the 3C cubic $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$ perovskite predominantly remains.

When adding DMAcI in excess with respect to lead to the $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$ perovskite system, it appears to act as a crystallizing agent that controls the speed at which the 2H–4H–6H–3C crystallization sequence takes place. In turn, this directly impacts the nucleation dynamics, crystallization rate and final crystal quality of the 3C perovskite film. These hexagonal polytypes may template the perovskite orientation due to their hexagonal crystal structure being similar to the 2H δ - FAPbI_3 ‘yellow’ phase, which also exhibits a strong diffraction peak at $2\theta \approx 11.8^\circ$. This offers a unique perovskite crystallization strategy, where the 2H to 3C crystallization sequence is controlled on a minute time frame, rather than seconds, as for the standard DMF/DMSO solvent system. Furthermore, we highlight

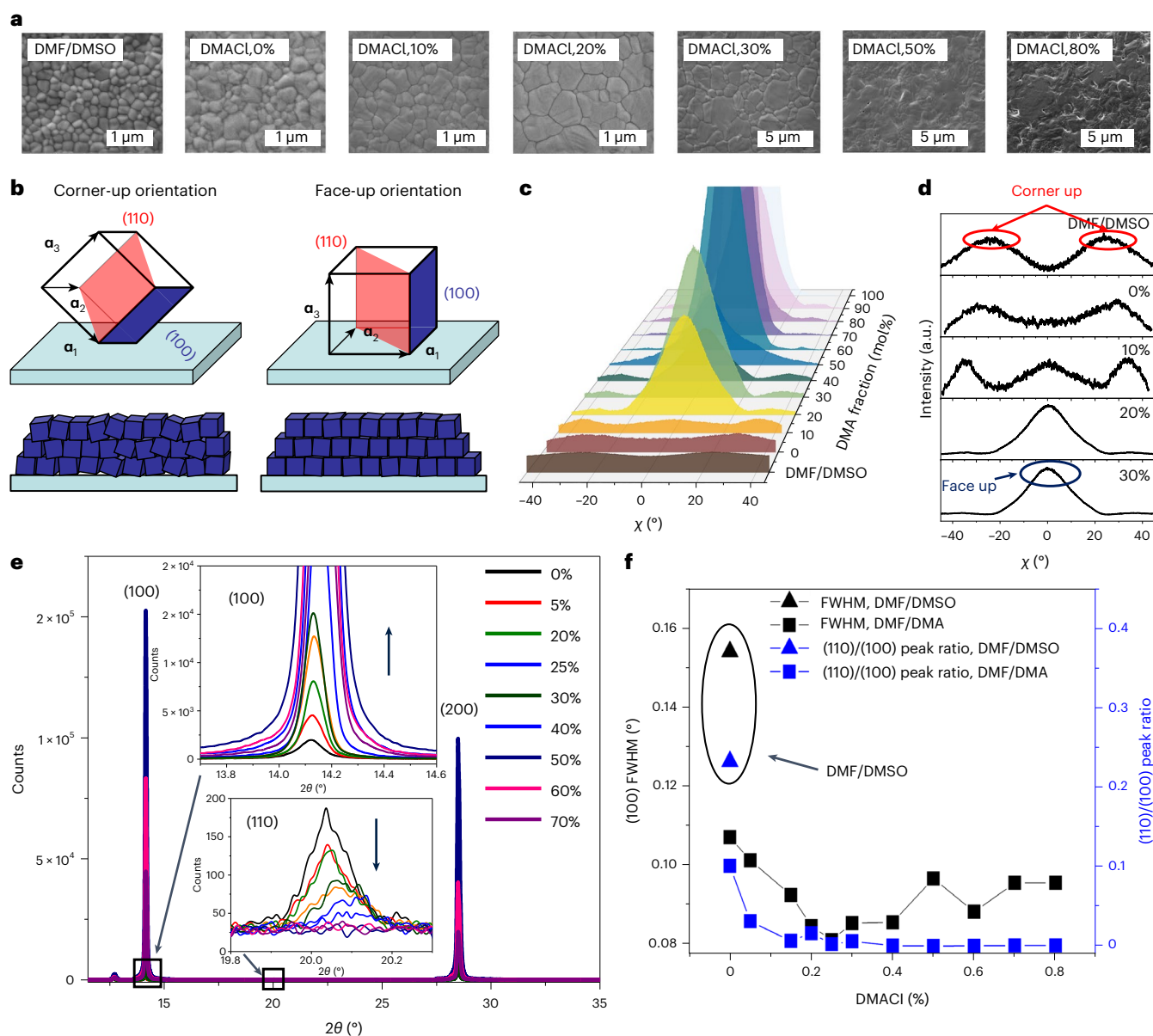


Fig. 3 | The impact of excess amounts of DMACI on the crystal quality, orientation and electronic disorder of the $\text{FA}_x\text{Cs}_{1-x}\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3$ perovskite. **a**, A series of SEM images of the crystal structure. **b**, A schematic illustration comparison between corner-up and face-up orientations with respect to the FTO substrate, where a_1 , a_2 , a_3 are lattice vectors. **c**, A series of 2D X-ray diffraction beta peak analyses of the (100) crystal plane for a series of $\text{DMA}_x(\text{FA}_{0.83}\text{Cs}_{0.17})\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3\text{Cl}_x$ perovskites, showing a corner-up orientation for the DMF/DMSO films and low DMACI content, but a predominant face-up orientation for films with 20% or more excess DMACI. **d**, A 2D X-ray diffraction beta peak analysis,

showing the intensity of the (100) peak as a function of the azimuth angle (χ) for a series of $\text{DMA}_x(\text{FA}_{0.83}\text{Cs}_{0.17})\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3\text{Cl}_x$ perovskite films. **e**, One-dimensional X-ray diffraction (Cu K α) patterns of $\text{DMA}_x(\text{FA}_{0.83}\text{Cs}_{0.17})\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3\text{Cl}_x$ perovskite thin films with various amounts of excess DMACI after a full annealing step. The inset figures show enlarged regions of the (100) and (110) X-ray diffraction peaks. **f**, A full-width at half-maximum (FWHM) of the (100) X-ray diffraction peak, measured with one-dimensional X-ray diffraction, and a (110) to (100) X-ray diffraction peak ratio for films with various DMA contents, compared to a film quenched by a DMF/DMSO solvent.

the fact that intermediate phases, induced with the addition of DMSO or DMA, are indispensable for avoiding large macroscopic cracks, which are otherwise found in films processed from neat DMF (Supplementary Fig. 12).

The intermediate phases induced by the DMSO- PbI_2 and/or $(\text{DMSO})_2\cdot\text{PbI}_2$ complexes disappear within the first few seconds of the annealing process, leading to rapid crystal formation. For our DMF/DMA route, we observe there is little to no hexagonal face-sharing DMAPbX_3 crystal structure remains when DMA is added in excess and the films are annealed at elevated temperatures ($>185^\circ\text{C}$; Supplementary Fig. 13). Films with higher DMACI content do require a longer

annealing time or higher temperatures to fully remove the DMAPbX_3 crystal structure, which is consistent with the postulation that the excess DMACI needs to be volatilized from the film for crystallization to occur. This is a similar requirement to the requirement when processing perovskite films from non-halide lead salts, such as lead acetate, or lead chloride with an excess of A-site cations.

Previous investigations of substituting FA/Cs or MA^+ for DMA^+ in lead halide perovskites have been reported^{48,49}, in contrast to the excess addition method we have developed in this work. In Supplementary Figs. 14–17, we investigate and discuss the differences between these two approaches, which result in perovskite films

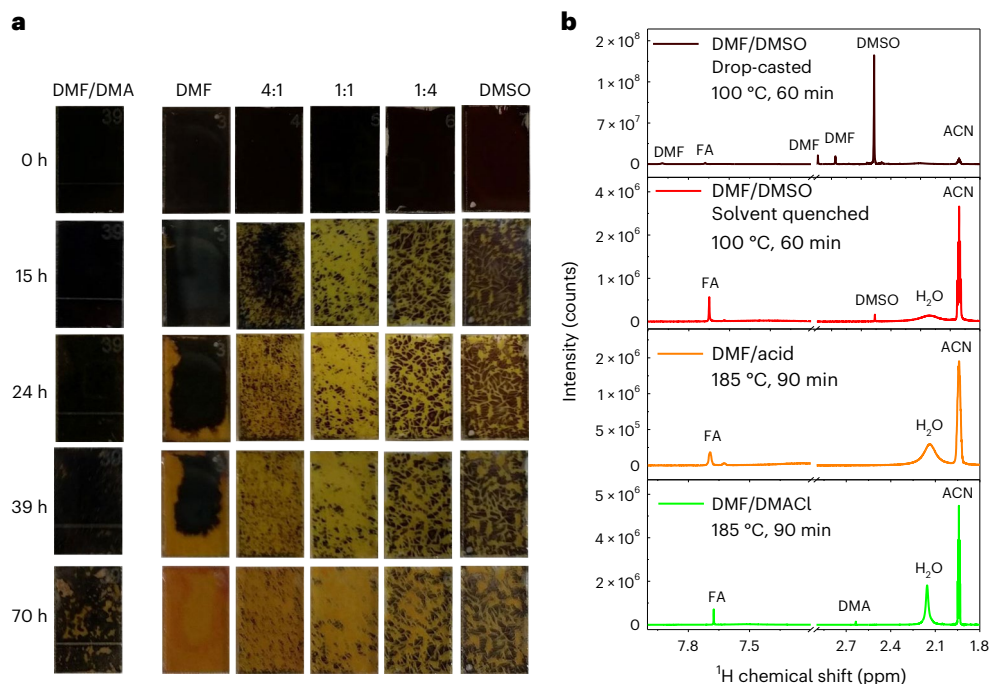


Fig. 4 | Impact of solvent and fabrication method on thin-film humidity stability. **a**, A series of photographs of spin-coated thin films fabricated with different ratios of DMF to DMSO, along with a comparison of the DMF/DMAcI method, at ~85% relative humidity. **b**, The ¹H NMR spectra of perovskite thin films

dispersed in ACN-d₃ (CD₃CN). NMR spectra of a perovskite thin film prepared with a DMF/DMSO drop-casted method (top), DMF/DMSO-solvent-quenched method (upper middle), DMF/acid method (lower middle) and DMF/DMAcI method (bottom).

with distinctly different crystallizations, morphologies and crystal qualities. Moreover, we have investigated the use of all DMAc cations and dimethylamine as an additive and have observed that all of these compounds can form hexagonal polytype intermediate phases (Supplementary Figs. 18–20). However, we found that in order to achieve an efficient device with a smooth perovskite morphology, DMAcI was the most suitable additive, resulting in DMA_x(FA_{0.83}Cs_{0.17})Pb(Br_{0.2}I_{0.8})Cl_x precursor phases. Notably, DMAcI allowed for faster removal of excess compounds at lower temperatures, and did not drastically narrow the bandgap of the perovskite, which is the case when adding DMAI/DMAcBr or HI/HBr in excess.

Due to the low dimensionality of the DMAcPbI₃ crystal structure, this compound appears to behave as an ionic conductor but suffers from poor electronic conductivity⁴⁶. These insulating properties may be the reason why the optimum annealing requires high temperatures along with long durations in order to drive out the face-sharing DMAcPbX₃ perovskite. However, we note that a small portion of DMA⁺ could potentially be incorporated into the cubic (FA_{0.83}Cs_{0.17})Pb(Br_yI_{1-y})₃ structure. Recent reports indicate that up to 30% of the caesium in CsPbI₃ can be substituted with DMA⁺, thus forming Cs_{0.7}DMA_{0.3}PbI₃ in a 3C corner-sharing cubic crystal structure²⁹. Similarly, it was shown that the Cs_xDMA_{1-x}PbI₃ perovskite forms only a pure-phase material with up to ~25% DMA⁺ substitution; above this point, the A-site cations begin to phase segregate³⁰. Hence, assuming similar incorporation fractions, 4–5% (25–30% of the 17% Cs) DMA⁺ cations may be incorporated into the final perovskite structure. Furthermore, we measured some residual DMA⁺ using TOF-SIMS (Supplementary Fig. 1). However, we did not observe a major change in absorption onset between the DMF/DMSO and DMF/DMAc fabrication methods after fully annealing the neat iodide perovskite film at high temperature (Supplementary Fig. 21). For the remainder of the discussion, we will assume that there is only residual DMA⁺ remaining in the final perovskite structure after crystallization.

Morphological and crystallographic properties

In Fig. 3, we compare the crystallographic and photophysical properties of FA_yCs_{1-y}Pb(I_xBr_{1-x})₃ perovskite films prepared with the DMF/DMSO and DMF/DMAcI routes. In Fig. 3c and Supplementary Figs. 22–24, we analyse the material quality using a two-dimensional (2D) X-ray diffraction system of a series of DMA_z(FA_yCs_{1-y})Pb(Br_xI_{1-x})₃Cl_z perovskite films. We observed that the DMF/DMSO perovskite films showed (100) X-ray diffraction peaks with an azimuth angle (χ) located at approximately $\chi = \pm 25^\circ$, and films prepared with DMF solvent with 10% or less DMAcI content resulted in (100) peaks located at approximately $\chi = \pm 35^\circ$, corresponding to crystallites whose (100) planes are not completely perpendicular to the substrate, while the (101) plane becomes more perpendicular to the substrate upon DMA⁺ addition, resulting in a corner-up perovskite film orientation. These films tend to be highly polycrystalline with low levels of texture. On the other hand, films with 20% or more DMAcI had a strong (100) X-ray diffraction peak located precisely at $\chi = 0^\circ$, corresponding to crystallites whose (100) planes lie perpendicular to the substrate, thus producing a face-up orientation. These findings coincide with our previous 2D X-ray diffraction analysis of the DMF/acid method, where films prepared with an aged solution with the addition of hydrohalic acids tend to orient their (100) peak to $\chi = 0^\circ$ (ref. 31). Films with a high DMA⁺ content have slower crystallization rates, allowing for a face-up orientation to occur, while films with low or no DMA⁺ content, such as the conventional DMF/DMSO films, crystallize at a much quicker rate, resulting in a large portion of crystallite with a corner-up orientation. We hypothesize that crystallites may preferentially orient with their (100) plane parallel to the substrate if enough time and thermal energy is available. Similarly, links between different MAPbI₃ perovskite fabrication methods prepared via various lead precursors, such as PbI₂, PbCl₂ or Pb(CH₃COO)₂, and their crystalline orientation were reported⁵⁰. Here, we expand on this knowledge by establishing links between the DMA-induced 2H–3C precursor phases, which govern the crystallization kinetics of the FA_yCs_{1-y}Pb(I_xBr_{1-x})₃

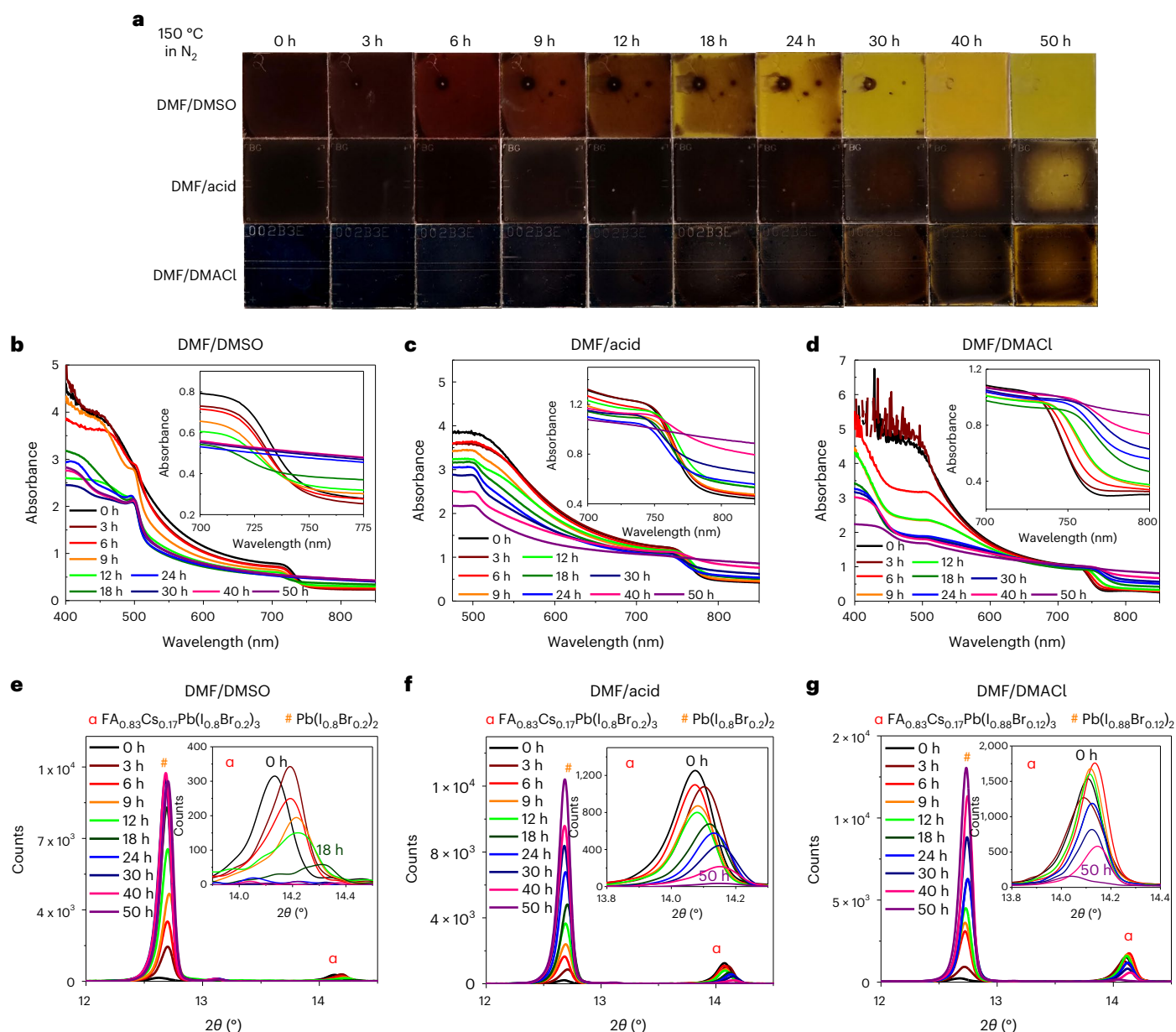


Fig. 5 | Thin-film thermal stability. **a**, Photographs of perovskite thin films when heated at 150 °C in a N_2 atmosphere, at various times. The top row shows $FA_{0.83}Cs_{0.17}Pb(I_{0.8}Br_{0.2})_3$ films fabricated with the standard DMF/DMSO fabrication method, the middle row shows $FA_{0.83}Cs_{0.17}Pb(I_{0.8}Br_{0.2})_3$ films fabricated with the DMF/acid fabrication method and the bottom row shows $DMAcI_{0.5}FA_{0.83}Cs_{0.17}Pb(I_{0.88}Br_{0.12})_3Cl_{0.5}$ films fabricated with the DMF/DMAcI fabrication method. **b–d**, UV–visible absorbance spectra corresponding to the films in **a**. The insets

in **b**, **c** and **d** show the enlarged UV–visible absorbance spectra near the optical bandgap of these films. **e–g**, X-ray diffraction patterns of the films in **a**, where # and α stand for the reflections originating from the PbX_2 (X is a mixture of I, Br and Cl) and cubic perovskite phases, respectively. The insets in **e**, **f** and **g** show the enlarged X-ray diffraction patterns, highlighting the evolution of the cubic perovskite phase during the thermal stressing course.

perovskite system in order to template the growth of the highly ordered face-up domains.

We show that the DMAcI mimics the behaviour of hydrohalic acid addition, controlling the grain sizes and the crystallographic properties of the thin films. By allowing a much greater time span for the 2H to 3C sequence to occur, we observe that the perovskite crystal can orient itself with a substantial (100) preferred orientation. As shown in Fig. 3e, by comparing X-ray diffraction patterns of perovskite thin films with roughly identical thicknesses and scan parameters, we observe an increase of almost two orders of magnitude ($\times 77$) in the number of counts between the DMF-processed neat $FA_{0.83}Cs_{0.17}Pb(Br_{0.2}I_{0.8})_3$ film and the $DMAcI_{0.5}(FA_{0.83}Cs_{0.17})Pb(Br_{0.2}I_{0.8})_3Cl_{0.5}$ -processed film. Concurrently, we observe the disappearance of the X-ray diffraction peak

associated with the (110) orientation located at $\sim 20.1^\circ$ for precursor solutions with over 40% DMAcI. Lastly, the full-width at half-maximum for the (100) peak was lowered by roughly a factor of two compared to the DMF/DMSO control, indicative of an increase in the crystal grain size and reduction in the microstrain of the perovskite material (Fig. 3f).

Heat and moisture stability and residual solvent

In Fig. 4, we show the impact of the composition of the perovskite processing solvent on the stability of perovskite thin films in a humid environment. A series of films prepared with various DMF-to-DMSO ratios was continually exposed to humid air with a relative humidity of approximately 85% at room temperature. We compared these films to our optimized, highly orientated and crystalline DMF/DMAcI-processed

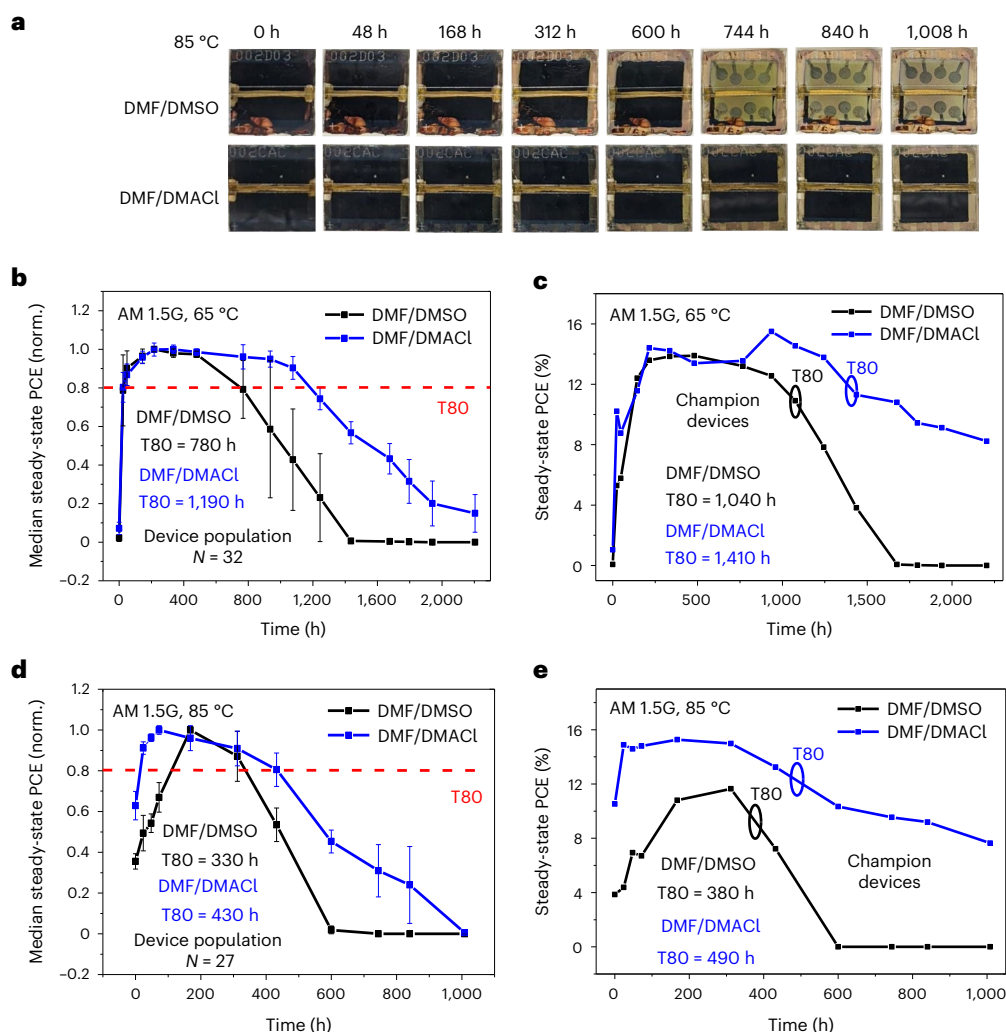


Fig. 6 | Stability comparison, aged in air under heat and light, of encapsulated perovskite solar cell devices prepared with DMF/DMSO or DMF/DMAcI fabrication method. a, Photographs of encapsulated DMF/DMSO (control) and DMF/DMAcI p-i-n devices taken at various ageing times under full-spectrum simulated AM 1.5G, 76 mW cm⁻² average irradiance at V_{oc} in air without a UV filter at 85 °C, using a Suntest XLS+ ageing box, which irradiates pulsed light. **b,c,** Evolution of normalized (norm.) steady-state PCE, measured at its peak efficiency, between two statistical populations (**b**) and champion devices (**c**)

of encapsulated devices of DMF/DMSO and DMF/DMAcI designs, aged under simulated sunlight at 6 °C; 32 devices were used in **b, d, e**. Evolution of normalized steady-state PCE, measured at its peak efficiency, between two statistical populations (**d**) and champion devices (**e**) of encapsulated devices of DMF/DMSO and DMF/DMAcI designs aged under simulated sunlight at 85 °C; 27 devices were used in **d**. The error bars were calculated using the median absolute deviation as a measure of statistical dispersion.

FA_{0.83}Cs_{0.17}Pb(I_{0.8}Br_{0.2})₃ perovskite films. We observed improved stability for the DMSO-free films and a considerable improvement in stability for the DMF/DMAcI-processed films.

DMSO is a common Lewis-base solvent known to complex and coordinate with lead halide (PbX₂). This ability allows intermediate precursor phases to occur and slow down the perovskite crystallization process and increase its processing window^{21,51}. However, the combination of strong molecular interaction with perovskite/PbX₂, along with a high boiling point, results in residual DMSO solvent remaining in the final perovskite film. Using nuclear magnetic resonance (NMR) spectroscopy, we analysed powders obtained from perovskite thin films that were dispersed in deuterated acetonitrile (ACN-d₃). By dispersing the perovskite powder in ACN-d₃, residual DMF or DMSO solvent can be detected. Figure 4b shows residual solvent peaks for both DMF (7.93, 2.89 and 2.77 ppm) and DMSO (2.51 ppm) for a thin film fabricated with a 50 µl drop-casted perovskite film, indicative that a 1 hour, 100 °C anneal is not sufficient to remove all solvent from a thick drop-casted film. Although a 500-nm-thick DMF/DMSO-solvent-quenched film did

not show any residual DMF, a DMSO peak could still be detected after a 1 hour, 100 °C annealing step. As shown in Supplementary Fig. 25, the DMF/DMAcI film can sustain a high-temperature annealing process (>170 °C), removing any residual solvent from the film. We suspect that the entrapped hygroscopic solvent is the cause for the reduced humidity stability of the DMF/DMSO-processed films^{26,52}.

In Fig. 5, we show the thermal degradation of various perovskite films prepared with different deposition techniques when heated at 150 °C in a N₂ atmosphere. We present the UV-visible absorption spectra in Fig. 5b–d, and the X-ray diffraction patterns in Fig. 5e–g, zoomed in at the 11–15° 2θ region of the corresponding FA_{0.83}Cs_{0.17}Pb(I_{0.8}Br_{0.2})₃ perovskite films. The DMF/DMSO perovskite film shows a faster rate of yellowing, which occurred within the first 24 hours of 150 °C heating, compared to both the DMF/acid- or DMF/DMAcI-processed films, which remained dark until 50 hours for narrow and wide perovskite compositions (Supplementary Fig. 26). We observed that the DMF/DMSO-processed perovskite films show a gradual decrease in the sharpness of the perovskite absorption onset and a complete disappearance

of absorption onset after only 24 hours of heating at 150 °C. On the other hand, the DMF/acid and the DMF/DMAcI fabrication routes showed roughly half the degradation rate, where the perovskite absorption onset fully disappeared after 40–50 h. Similarly, we observed the complete loss of the perovskite (100) peak, located at $2\theta \approx 14^\circ$, after 18 hours of heating for the DMF/DMSO route and 50 hours for both the DMF/acid and DMF/DMAcI route films. Correspondingly, we observe a rise of the $\text{Pb}(\text{Br}_{x-1}\text{I}_x)_2$ peak located around $2\theta \approx 12.7^\circ$. In Supplementary Fig. 27, we present the evolution of the normalized intensity ratio between the lead halide and perovskite (100) X-ray diffraction peaks versus thermal ageing time. To further assess and quantify the thermal stability of these films, we performed thermogravimetric analysis directly on perovskite-coated glass substrates to account for the variation in morphology, topography and crystal orientation of films produced via these different fabrication methods. In Supplementary Fig. 28, we show a 14 °C difference, from 265 to 279 °C, between the peaks of the first derivative of DMF/DMSO and DMF/acid films. Further details and results are provided in Supplementary Figs. 28–30. Lastly, we conducted an in situ thermal degradation X-ray diffraction study of a series of films, with/without DMSO and/or DMAcI, stressed at 130 °C in an air atmosphere with ~33% relative humidity (Supplementary Fig. 31).

To assess the performance of these different processed perovskite absorber layers in complete solar cells, we fabricated both negative–intrinsic–positive (n–i–p) and p–i–n device configurations. The device structures were as follows: fluorinated tin oxide (FTO)/ SnO_2 compact layer (CL)/ SnO_2 chemical bath deposition (CBD)/KCl/perovskite/phenethylammonium iodide (PEAI)/2,2',7,7'-tetrakis(*N,N*-di(4-methoxyphenyl)amino)-9,9'-spirobifluorene (spiro-OMeTAD)/Au for the n–i–p cells; and FTO/poly(bis(4-phenyl)(2,4,6-trimethylphenyl)) amine (PTAA)/ Al_2O_3 nanoparticles/perovskite/LiBr/ C_{70} /zirconium acetylacetonate ($\text{Zr}(\text{acac})_4$)/polyethylenimine ethoxylated (PEIE)/Au for the p–i–n cells. The devices fabricated via the DMF/DMAcI route reached 20.2% PCE (19.9% steady state) in the n–i–p configuration and 16.7% PCE (16.1% steady state) in the p–i–n configuration, which is comparable to the DMF/DMSO-processed devices, which achieved 18.8% PCE (19.2% steady state) in the n–i–p configuration and 16.2% PCE (14.8% steady state) in the p–i–n configuration (Supplementary Fig. 32). The difference between the integrated short-circuit current density, J_{sc} from the EQE and the J_{sc} measured under air mass 1.5 global (AM 1.5G) solar simulation is within 5% (Supplementary Fig. 33).

Although the n–i–p, spiro-OMeTAD-based devices tend to outperform the p–i–n devices in terms of PCE, their operational stability suffers due to hole transporting material (HTM) instabilities linked to hygroscopic and volatile dopants^{53,54}. In order to assess the long-term stability of these two distinct crystallization routes in complete photovoltaic devices, we aged two populations of p–i–n devices (DMF/DMSO and DMF/DMAcI) under simulated sunlight at 65 °C for over 2,000 hours, and a separate set of populations at 85 °C for over 1,000 hours (Fig. 6a–e). We employed a glass/glass encapsulation technique, sealed under a N_2 atmosphere with a UV-cured epoxy resin.

Accelerated ageing

In contrast to the n–i–p cells, these p–i–n cells do not exhibit a sharp early degradation, known as the 'burn-in' period⁷. By contrast, we saw a drastic gain in PCE over the first 200 hours of light and temperature exposure. We used this peak PCE to estimate the T80 lifetime of encapsulated cells, under full-spectrum sunlight illumination, at 65 °C and open-circuit conditions, resulting in a T80 lifetime of 1,040 (780) hours for the champion (median) control DMF/DMSO device and 1,410 (1,190) hours for the champion (median) DMF/DMAcI device⁷. Furthermore, for ageing past 1,600 hours, we observed that the control devices have all completely died, consistent with their bleached visual appearance (Fig. 6a–c), whereas the DMF/DMAcI-processed devices sustain on average 50% of their peak performance at 1,600 hours, with the hero cell sustaining >70% of its peak performance at this time. We

aged a sizeable device population of $N = 106$ current density–voltage (J–V) curve and $N = 32$ (stabilized) devices to establish a statistical significance between these two designs (Fig. 5b,c and Supplementary Fig. 34). We also conducted the same full-illumination degradation study at a higher temperature of 85 °C, where the DMF/DMSO devices obtained a champion (median) T80 of 380 (330) hours, while the DMF/DMAcI devices obtained a champion (median) T80 of 490 (430) hours for a population size of $N = 72$ (J–V) and $N = 27$ (stabilized; Fig. 5d,e and Supplementary Fig. 35). Hence, the improved stability of the perovskite absorber layer processed via the DMF/DMAcI route has resulted in a substantial increase in the overall photovoltaic device stability under elevated temperature light soaking. Incidentally, from these results, we can estimate a thermal degradation acceleration factor of roughly 1.7-fold per 10 °C increment for these devices. This can be useful for comparing the relative stability of different materials and devices stressed at different temperatures and is close to the 2-fold acceleration in degradation per 10 °C increment, which is expected for the degradation of silicon photovoltaic cells. We note here that we age our devices at open-circuit conditions, which is expected to accelerate the degradation compared to measuring it at the maximum power point⁵⁵.

Outlook

Here, we have presented a method of accurately controlling the precursor phase evolution involved from the 2H to 3C crystallization sequence of the $\text{FA}_{x-1}\text{Cs}_1\text{Pb}(\text{Br}_{x-1}\text{I}_x)_3$ perovskite system via the addition of DMAcI. This allowed us to remove DMSO, a hygroscopic solvent, from the precursor solution, which is commonly used to form $\text{DMSO}\cdot\text{PbI}_2$ and $(\text{DMSO})_2\cdot\text{PbI}_2$ complexes to control the intermediate phases during the crystallization process. This led to fewer structural defects, which resulted in higher overall perovskite stability on many fronts, including thermal, humidity and light stabilities. We expect future improvements to be achieved by further understanding these intermediate phases and the mechanism by which they impact the crystallization kinetics and grain orientation of perovskite films. This insight may allow these intermediate phases to act as both key indicators and an avenue for improved process control to reduce structural defects for higher stability.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-022-01399-8>.

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Methods

Fabrication of p–i–n perovskite devices comprised of a glass/FTO/PTAA/ Al_2O_3 nanoparticles/perovskite/LiBr/ C_{70} /Zr(acac) $_4$ /Au architecture

Substrate preparation. Devices were fabricated on FTO-coated glass (Pilkington, 15 ohms per square ($\Omega \text{ sq}^{-1}$)). Initially, FTO was removed at specific regions where the anode contact would be deposited. This FTO etching was done using 2 M HCl and zinc powder. Substrates were then cleaned with Hellmanex detergent and rinsed with water. Finally, the substrates were cleaned sequentially in acetone and isopropyl alcohol (IPA) and dried with a nitrogen gun.

PTAA layer fabrication. A PTAA precursor solution was prepared by dissolving 1.5 mg ml^{-1} PTAA (Xi'an Polymer) in 1:1 anhydrous chloroform/chlorobenzene. We spin-coated this solution dynamically in a nitrogen-filled glove box at 2,000 r.p.m. for 20 s and annealed the substrate at 150 °C for 10 min in air.

Al_2O_3 nanoparticle layer fabrication. A precursor solution of alumina (Al_2O_3) nanoparticles (<50 nm) was prepared by dispersing IPA in a 1:150 v/v ratio of Al_2O_3 nanoparticles to IPA. Then 50 μl of this dispersion was dynamically spin-coated onto the PTAA layer at 4,000 r.p.m. for 30 s. A subsequent anneal of at least 5 min at 85 °C was done.

DMF/DMSO perovskite precursor solution preparation. For the $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$, the FA/Cs perovskite solutions were prepared by dissolving the precursor salts—171.3 mg FAI (99.99%, GreatCellSolar), 53.0 mg CsI (99.9%, Alfa Aesar), 132.1 mg PbBr_2 (98%, Alfa Aesar) and 387.2 mg PbI_2 (99.99% trace metals and a purity of >98.0% (iodometric titration), TCI)—in 1 ml anhydrous *N,N*-DMF/DMSO in a 4:1 v/v solvent ratio to obtain a stoichiometric solution with the desired composition. For fine-tuning the Br/I ratio, we varied the ratio of PbI_2 to PbBr_2 to obtain a precursor solution with the desired halide molecular ratio.

DMF/DMSO perovskite layer preparation. The DMF/DMSO perovskite precursor solution was spin-coated in a nitrogen-filled glove box using a two-step programme. After spreading 50 μl of the perovskite solution onto the substrate, the first step was programmed to spin at 1,000 r.p.m. for 10 s. The second step was programmed at 6,000 r.p.m. for 20 s. Then, 10 s prior to the end of the second step, 100 μl of ethyl acetate solvent was dropped on the film. The films were immediately placed on a hotplate in N_2 at 100 °C for 20 min.

DMF/acid perovskite precursor solution preparation. For the $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$, the FA/Cs perovskite solutions were prepared by dissolving the precursor salts—171.3 mg FAI (99.99%, GreatCellSolar), 53.0 mg CsI (99.9%, Alfa Aesar), 132.1 mg PbBr_2 (98%, Alfa Aesar) and 387.2 mg PbI_2 (99.99% trace metals and a purity of >98.0% (T), TCI)—in 1 ml anhydrous *N,N*-DMF to obtain a stoichiometric solution with the desired composition. Some 85.6 μl hydroiodic acid (HI; 57 wt% in H_2O , distilled, stabilized, 99.95%, Sigma-Aldrich) and 18.4 μl hydrobromic acid (HBr; 48 wt% in H_2O) was added for every 1 ml DMF. After the addition of the acids, the perovskite precursor solution was aged for 48 h in a nitrogen atmosphere. For fine-tuning the Br/I ratio, we varied the ratio of PbI_2 to PbBr_2 , along with the HI to HBr ratio, to obtain a precursor solution with the desired halide molecular ratio.

DMF/acid perovskite layer preparation. The DMF/acid perovskite precursor solution was aged for 48 h under a N_2 atmosphere and then spin-coated in a nitrogen-filled glove box at 2,000 r.p.m. for 45 s with a 1,000 r.p.m. s^{-1} ramp rate. The films were dried inside the N_2 glove box on a hotplate at a temperature of 70 °C for 1 min. The films were then annealed in an oven in an air atmosphere at 185 °C for 90 min. During this annealing process, the samples were covered with a large glass container to prevent dust contamination.

DMF/DMAcI perovskite precursor solution preparation. For the $\text{DMAc}_{0.83}\text{FA}_{0.17}\text{Pb}(\text{Br}_{0.12}\text{I}_{0.88})_3\text{Cl}_x$ perovskite solution, we dissolved the precursor salts—199.8 mg FAI, 61.8 mg CsI, $X \text{ mol\%}$ DMAcI (for example, 34.2 mg for $X = 0.3$), 92.5 mg PbBr_2 and 529.2 mg PbI_2 —in 1 ml anhydrous *N,N*-DMF to obtain a stoichiometric solution with the desired composition. The precursor solution was prepared using the following precursor salts: FAI (99.99%, GreatCellSolar), CsI (99.9%, Alfa Aesar), PbI_2 (99.99% trace metals and a purity of >98.0% (T), TCI), PbBr_2 (98%, Alfa Aesar) and DMAcI ($\geq 98.0\%$, Alfa Aesar). For fine-tuning the Br/I ratio, we varied the ratio of PbI_2 to PbBr_2 to obtain a precursor solution with the desired halide molecular ratio. Similarly, we fine-tuned the DMAcI concentration by varying the precursor solution to obtain the desired molecular ratio.

DMF/DMAcI perovskite layer deposition. The DMF/DMAcI perovskite precursor solution was spin-coated in a nitrogen-filled glove box using a two-step programme. After spreading 50 μl of the perovskite solution onto the substrate, the first step was programmed to spin at 1,000 r.p.m. for 7 s; the second step was programmed at 4,000 r.p.m. for 20 s. Then, 5 s after the start of the second step, 100 μl of a 3:1 α,α,α -trifluorotoluene/ethyl-acetate solvent mixture was dropped on the film to quench the growth of perovskite. The films were immediately placed on a hotplate in N_2 at 100 °C for 5 min. This step is designed to remove the solvent; however, at this point, the perovskite remains in its yellow intermediate phase. A subsequent annealing step on a 170 °C preheated hotplate in air for 10 min allows for the 2H to 3C sequence to take place, where the films turn from yellow to black. A final annealing step takes place in a 175 °C preheated oven in air for 90 min to fully crystallize the perovskite. In the oven, a large glass cover lid is placed over the samples to prevent dust particles and airflow from affecting the annealing.

FABr layer deposition. A FABr precursor solution was prepared by dissolving 2 mg ml^{-1} FABr (99.99%, GreatCellSolar) in IPA. After the full crystallization of the perovskite, we spin-coated this solution dynamically in a nitrogen-filled glove box at 4,000 r.p.m. for 20 s, and annealed the substrate at 100 °C for 5 min in N_2 . The FABr solution was dropped within the first 5 s of spin-coating. The FABr was deposited on all p–i–n devices of various designs.

LiBr layer deposition. A LiBr precursor solution was prepared by dissolving 0.15 mg ml^{-1} LiBr (Sigma) in IPA. We spin-coated this solution dynamically in a nitrogen-filled glove box at 4,000 r.p.m. for 20 s and annealed the substrate at 85 °C for 10 min in N_2 .

C_{70} layer fabrication. A C_{70} fullerene precursor solution was prepared by dissolving 24 mg ml^{-1} C_{70} (Solenne) in dichlorobenzene. We spin-coated this solution statically in a nitrogen-filled glove box at 800 r.p.m. for 40 s, on a preheated substrate at 85 °C. We annealed the substrate at 85 °C for 10 min in N_2 .

Zr(acac) $_4$ /PEIE layer fabrication. A Zr(acac) $_4$ /PEIE precursor solution was prepared by dissolving 2 mg ml^{-1} Zr(acac) $_4$ (Sigma, 97%) in IPA and adding 0.1% (v/v) PEIE (80% ethoxylated solution, 37 wt% in H_2O , Sigma) to the solution. We spin-coated this solution dynamically in a nitrogen-filled glove box at 4,000 r.p.m. for 20 s.

Fabrication of n–i–p perovskite devices comprised of a glass/FTO/ SnO_2 CL/ SnO_2 CBD/KCl/perovskite/PEAl/spiro-OMeTAD/Au architecture

Substrate preparation. Devices were fabricated using FTO-coated glass (Tec7, Pilkington) as the transparent electrode. FTO was etched with 2 M HCl and zinc powder to obtain the required electrode pattern. Following this, a cleaning procedure consisting of a series of sonication steps were applied: first, 10 min in Decon (5% in deionized water), followed by 10 min in acetone and finally a 10 min sonication in IPA.

SnO₂ CL. The substrates were plasma cleaned (Pico, Diener Electronic) for 10 min. In the meantime, a 17.5 mg ml⁻¹ solution of SnCl₂·2H₂O (Alfa Aesar, 99.995%) in IPA was prepared. Statically, 200 ml of the SnCl₂ solution was deposited at 3,000 r.p.m. (200 r.p.m. s⁻¹ acceleration) for 25 s in ambient atmosphere (between 30 and 50% relative humidity, 20 to 25 °C). The substrates were annealed at 100 °C until all substrates were spun and brought onto a sintering hotplate. They were heated with the lid closed and no gas flow attached for 1 h at 180 °C.

SnO₂ CBD. We first prepare a bath solution for 32 substrates (28 mm × 28 mm each). The solution was prepared with 200 ml deionized water, 2.5 g urea (Sigma-Aldrich), 50 µl thioglycolic acid (Sigma-Aldrich, 98%), 2.5 ml HCl (99.999%, Alfa Aesar, no. 087617-AK, CAS no. 7647-01-0) and 540 mg SnCl₂ (≥99.99%, Sigma-Aldrich, no. 431508, 50 g, CAS no. 10025-69-1). After adding each component, the solution was stirred with a stir bar for a few minutes. We then added the substrates into a glass beaker, filled it with the chemical bath solution and covered it with a glass cover. The substrates were then heated in a ventilated oven at 90 °C for 4 h. After the heating process, the chemical bath was drained, and the glass beaker was quickly filled with deionized water. Afterward, the substrates were sonicated for 5 min in deionized water, dried with a N₂ gun and then rinsed with IPA for 5 min. Finally, the substrates were heated at 180 °C on a sintering hotplate, as previously described, for 1 h. The substrates were subjected to a UV–ozone treatment for 15 min before use.

KCl treatment. We used 20 mM KCl solution in deionized water. We statically spin-coated 200 µl of the solution at 3,000 r.p.m. (2,000 acceleration) for 30 s in ambient atmosphere onto the substrate. The substrate was annealed on a hotplate at 100 °C for 10 min.

DMF/DMSO perovskite preparation. For the FA_{0.83}CS_{0.17}Pb(I_{0.88}Br_{0.12})₃ perovskite, 827.86 mg FAI (99.999%, Dyenamo), 256.17 mg CsI (99.998%, Alfa Aesar), 2,192.56 mg PbI₂ (99.999% Alfa Aesar) and 382.16 mg PbBr₂ (98%, Alfa Aesar) were dissolved in 4 ml of 4:1 DMF/DMSO. The mixture was stirred at room temperature in the glove box for 1 to 3 hours before deposition. Some 70 µl of the solution was spread onto the substrate statically and spin-coated for 10 s at 1,000 r.p.m. (200 r.p.m. s⁻¹ acceleration) and 35 s at 6,000 r.p.m. (2,000 r.p.m. s⁻¹ acceleration), and then solvent quenched with 150 µl anisole for 10 s before the end. The layers were annealed at 100 °C for 15 min. Spinning and heating for the control device were done in a dry box at 20% relative humidity.

DMF/DMAcI perovskite preparation. For the DMAcI_{0.3}FA_{0.83}CS_{0.17}Pb(I_{0.88}Br_{0.12})₃ perovskite, 136.99 mg DMAcI (Sigma-Aldrich), 799.32 mg FAI (99.999%, Dyenamo), 247.34 mg CsI (99.998%, Alfa Aesar), 2,117.0 mg PbI₂ (99.999%, Alfa Aesar) and 369.9 mg PbBr₂ (98%, Alfa Aesar) were dissolved in 4 ml DMF. The mixture was stirred at room temperature in the glove box for 1 to 3 hours before deposition. Some 70 µl of the solution was spread onto the substrate statically and spin-coated for 10 s at 1,000 r.p.m. (1,000 r.p.m. s⁻¹ acceleration) and 30 s at 4,000 r.p.m. (4,000 r.p.m. s⁻¹ acceleration), and then solvent quenched with 150 µl α,α,α-trifluorotoluene at 17 s into the programme. The layers were annealed at 100 °C for 10 min and directly afterward at 170 °C. The 30 mol% DMAcI/FA_{0.83}CS_{0.17}Pb(I_{0.88}Br_{0.12})₃ perovskite was spun in a dry box at 20% relative humidity at -23–25 °C. The heating for the 170 °C step was performed in ambient conditions (30 to 55% relative humidity).

PEAI layer deposition. The PEAi precursor solution was prepared by dissolving 1.25 mg ml⁻¹ PEAi (99.99%, GreatCellSolar) in IPA. After the full crystallization of the perovskite, we spin-coated this solution dynamically in a dry, air-filled glove box at 5,000 r.p.m. for 20 s, and annealed the substrate at 100 °C for 5 min in a dry air atmosphere. The PEAi solution was dropped within the first 5 s of spin-coating.

Spiro-OMeTAD preparation. The hole transfer materials were deposited by preparing spiro-OMeTAD in chlorobenzene (90 mg ml⁻¹, Lumtech) and mixing with 35.5 µl 4-*tert*-butylpyridine, 23 µl Li-bis(trifluoromethanesulfonyl)imide (520 mg in 1 ml of acetonitrile) and 5 µl tris(2-(1*H*-pyrazol-1-yl)-4-*tert*-butylpyridine)-cobalt(III)tris(bis(trifluoromethylsulfonyl)imide) (FK209, Lumtech; 180 mg in 1 ml of acetonitrile). The spiro-OMeTAD layer was deposited at 4,000 r.p.m. for 25 s with 2,000 r.p.m. s⁻¹ acceleration in a dry box at 20% relative humidity.

Metal electrode layer fabrication. An 80 nm gold electrode was thermally evaporated under a vacuum of -10⁻⁶ torr, at a rate of -0.2 nm s⁻¹ or 0.01–0.04 nm s⁻¹.

Encapsulation. A glass/glass encapsulation was created by sealing the edge of the glass with a UV-cured epoxy (LT-U001 UV-Glue, Lumtec). The epoxy was cured for 30 min with a UV lamp containing four 9 W bulbs under a N₂ atmosphere.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets used in this work are available in the Oxford University Research Archive repository.

Acknowledgements

The research leading to these results has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement no. 763977 of the PerTPV project and the innovation programme under Marie Skłodowska-Curie grant agreement no. 764787. We acknowledge financial support from the Engineering and Physical Sciences Research Council (UK), grant EP/S004947/1. We also acknowledge the financial support from the Australian Research Council Centre of Excellence in Exciton Science (ACEx:CE170100026). D.P.M. acknowledges financial support from the Australian Centre for Advanced Photovoltaics, the Australian Renewable Energy Agency and the Marie Skłodowska-Curie grant agreement SAMA no. 101029896. The work by S.P.H. and L.T.S. was supported by the De-Risking Halide Perovskite Solar Cells programme of the National Center for Photovoltaics, funded by the US Department of Energy, Office of Energy Efficiency and Renewable Energy, Solar Energy Technologies Office under US Department of Energy contract no. DE-AC36-08GO28308 with Alliance for Sustainable Energy, LLC, the Manager and Operator of the National Renewable Energy Laboratory. Work by J.J.B. was supported by the Office of Naval Research. The views expressed in the article do not necessarily represent the views of the US Department of Energy or the US Government. We acknowledge F. Vollrath and the Oxford Silk Group for their help and equipment. Use of the Stanford Synchrotron Radiation Lightsource, SLAC National Accelerator Laboratory, was supported by the US Department of Energy, Office of Basic Energy Sciences under contract no. DE-AC02-76SF00515. We also acknowledge the Monash X-ray Platform.

Author contributions

D.P.M. contributed to the conceptualization, investigation, methodology, analysis and original draft. P.H., S.O.F., S.P.H. and L.T.S. contributed to the investigation, methodology and analysis. J.M.B. contributed to the investigation. S.M. contributed to the analysis. S.S. contributed to the investigation and methodology. N.H. and J.L. contributed to the investigation. M.B.J. and J.J.B. supervised the research. U.B. contributed to funding acquisition, resources and supervision. H.J.S. contributed to the conceptualization, funding

acquisition, resources, supervision and the original draft. All authors contributed to the writing of the paper.

Competing interests

H.J.S. is founder and CSO of Oxford Photovoltaics Ltd. All other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41563-022-01399-8>.

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Peer review information *Nature Materials* thanks Aram Amassian, Nam-Gyu Park and Eva Unger for their contribution to the peer review of this work.

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☒ Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	Supplementary Information page 37
Method used to determine the device area	☒ Yes ☐ No	Supplementary Information page 37

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Supplementary Information page 32
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes ☐ No	Supplementary Information page 37
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes ☐ No	Supplementary Information page 38
Protocol for preconditioning of the device before its characterization	☒ Yes ☐ No	Supplementary Information page 37
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☒ Yes ☐ No	Supplementary Information page 37

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	We observe hysteretic behaviour, hence we show stabilized PCE for 30 s.
Related experimental data	☒ Yes ☐ No	Supplementary Information page 32

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	Fig. 1f-g, Supplementary Figure S5, Supplementary Figure S6
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	Supplementary Figure S33
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	 N/A

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Supplementary Information page 37
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Supplementary Information page 37

Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	Supplementary Information page 37
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Supplementary Information page 37
Variation of the measured short-circuit current density with the mask/aperture area	☒ Yes ☐ No	Supplementary Figure S34
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☐ Yes ☒ No	 N/A
A copy of any certificate(s) <i>Provide in Supplementary Information</i>	☐ Yes ☒ No	 N/A
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Manuscript page 17, Fig. 6
Statistical analysis of the device performance	☒ Yes ☐ No	Fig. 6
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions <i>For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature</i>	☒ Yes ☐ No	Fig. 6, Supplementary Figure S34, Supplementary Information page 37

✕ Further reading

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A number of international committees develop industry standards on the characterization of photovoltaic technologies (for example [ASTM-E44](#) and [IEC-TC 82](#)), which can provide guidance for academic research.

